

Vipul Harsh

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Interests	Current interests: reliability of AI agents, failure diagnosis in networked systems, network architectures Broad interests: networked and distributed systems, systems for AI	
Experience	Postdoctoral researcher, Conviva, Foster City, California	July 2024 - Ongoing
	Visiting researcher, CMU Advisor: Vyas Sekar	
	Research intern, VMware Research, Palo Alto, California (remote) Advisors: Sujata Banerjee , Radhika N. Mysore	June 2020 - Sept. 2020
Education	Software engineering intern, Google, Sunnyvale, California Advisors: Gautam Kumar , Nandita Dukkupati	May 2018 - August 2018
	University of Illinois at Urbana-Champaign Ph.D., Computer Science Advisor: P. Brighten Godfrey	2017 - 2024
	University of Illinois at Urbana-Champaign M.S., Computer Science Advisor: Laxmikant Kale	2015 - 2017
	Indian Institute of Technology, Bombay B.Tech. (Honors), Computer Science and Engineering	2011 - 2015
	Publications	
MoCE: A Mixture of Context Aware Experts Framework for Troubleshooting Internet-scale Services Vipul Harsh , Sayan Sinha, Henry Milner, B. Aditya Prakash, Vyas Sekar, Hui Zhang NSDI 2026		
Starfish: A Topology-Routing Co-Design for Small-Scale Data Centers Anchengcheng Zhou, Vipul Harsh , Sangeetha A. Jyothi, Maria Apostolaki, P. Brighten Godfrey NSDI 2026		
Automatically Surfacing Opportunities for Improvements In Internet-Scale Applications Vipul Harsh , Sayan Sinha, Henry Milner, Haijie Wu, B. Aditya Prakash, Vyas Sekar, Hui Zhang HotNets 2025		
TraceWeaver: Distributed Request Tracing for Microservices Without Application Modification Sachin Ashok, Vipul Harsh , P. Brighten Godfrey, Radhika Mittal, Srinivasan Parthasarathy, Larisa Shwartz SIGCOMM 2024		
Murphy: Performance Diagnosis of Distributed Cloud Applications Vipul Harsh , Wenxuan Zhou, Sachin Ashok, Radhika N. Mysore, P. Brighten Godfrey, Sujata Banerjee SIGCOMM 2023		
• VMware's blog post about product adoption: https://shorturl.at/efvT2		
Flock: Accurate Datacenter Fault Localization at Scale Vipul Harsh , Tong Meng, Kapil Agrawal, P. Brighten Godfrey CoNEXT 2023		
Optimal Round and Sample-Size Complexity for Partitioning in Parallel Sorting Wentao Yang*, Vipul Harsh* , Edgar Solomonik (*: equal contribution) SPAA 2023		
Spineless Datacenters Vipul Harsh , Sangeetha A. Jyothi, P. Brighten Godfrey HotNets 2020		

Histogram Sort with Sampling

Vipul Harsh, Laxmikant Kale, Edgar Solomonik
SPAA 2019

Manuscripts in preparation	There's Waldo: Localizing failures among symmetric components Vipul Harsh, Rahul Bothra, P. Brighten Godfrey
	Building Reliable Troubleshooting Agents for Internet-scale Services Using Structured Creativity Sayan Sinha, Vipul Harsh, B. Aditya Prakash, Vyas Sekar, Hui Zhang
Patents	• On-demand Network Incident Graph Generation (US Patent App. 18/094,378) Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee • Network Incident Root-Cause Analysis (US Patent App. 18/094,379) Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee • Providing Explanation of Network Incident Root Causes (US Patent App. 18/094,380) Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee
Awards	• Awarded a travel grant for NSDI 2026 • Selected for the NSF-NetS early career workshop, January 2025 at Alexandria, VA, USA • Represented IIT Bombay at the ACM ICPC World Finals 2015. Highest ranked team from India • Ranked 49 in IIT-JEE 2011, amongst 500,000 candidates • Rank 1 in International Mathematics Olympiad, 2009 conducted by Science Olympiad Foundation
Internships	• Research Internship, UC Berkeley June 2019 - August 2019 Distributed garbage collection for actor-based systems • Research Internship, Georgia Tech May 2014 - July 2014 Large scale simulations for polydisperse hydrodynamic particle systems • Research Internship, LaBRI, France May 2013 - July 2013 Algorithms for computing coverability sets of Petri Nets
Teaching	• Teaching Assistant, Cloud Networking, UIUC (Spring 2024) • Teaching Assistant, Probability and Statistics for Computer Science, UIUC (Spring 2022) • Teaching Assistant, Discrete Mathematics, IIT Bombay • Teaching Assistant, GPU Programming and Applications Workshop (GPA), IIT Bombay
Talks	• NSDI 2026, Renton, Washington, USA May 2026 MoCE: A Mixture of Context Aware Experts Framework for Troubleshooting Internet-scale Services • IISc Bengaluru, IIT Bombay, IIT Delhi, IIT Chennai, IIIT Hyderabad, Microsoft Research Bengaluru, India February-March 2026 Abstractions for Root Cause Analysis (RCA) in Networked Systems • HotNets 2025, College Park, Maryland, USA November 2025 Automatically Surfacing Opportunities for Improvements In Internet-Scale Applications • CMU, Pittsburgh, Pennsylvania, USA November 2025 Abstractions for high-coverage, extensible and scalable root cause analysis (RCA) Hosted by Prof. Vyas Sekar • MIT, Cambridge, Massachusetts, USA July 2024 Failure Diagnosis in Networked systems Hosted by Prof. Christina Delimitrou • Conviva, Foster city, California, USA March 2024 Failure Diagnosis in Networked systems Hosted by Prof. Vyas Sekar • CoNEXT 2023, Paris, France December 2023 Flock: Accurate Network Fault Localization at Scale

- **SIGCOMM 2023**, New York City, USA September 2023
Murphy: Performance Diagnosis of Distributed Cloud Applications
- **SPAA 2023**, Orlando, Florida, USA June 2023
Optimal Round and Sample-Size Complexity for Partitioning in Parallel Sorting
- **EnvoyCon at KubeCon**, Detroit, Michigan, USA October 2022
Distributed Tracing without the pain
- **VMware Research**, Palo Alto, California, USA August 2022
Murphy: Performance diagnosis of Distributed Cloud Applications
- **VMware RADIO**, San Francisco, California, USA May 2022
Murphy: Performance diagnosis of Distributed Cloud Applications
- **HotNets 2020**, Chicago, Illinois, USA November 2020
Spineless Data Centers
- **VMware**, Palo Alto, California, USA February 2020
Fast and accurate datacenter fault localization
- **Google**, Sunnyvale, California, USA August 2019
Fast and accurate datacenter fault localization
- **SPAA 2019**, Phoenix, Arizona, USA June 2019
Histogram sort with sampling
- **Charm++ Workshop**, Urbana, Illinois, USA April 2017
Histogram sort with sampling

References
(on request)

- **Vyas Sekar**, CMU, Conviva
- **Brighten Godfrey**, UIUC
- **Sujata Banerjee**, Microsoft Research
- **Edgar Solomonik**, UIUC

Service

- Program Committee, CoNEXT 2025